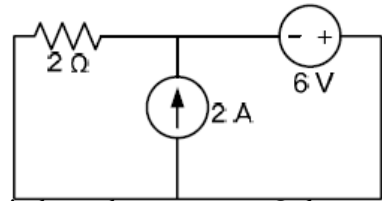


Quiz-1

For the first 5 questions, darken the bubble corresponding to the correct answer. -0.5 for wrong answers

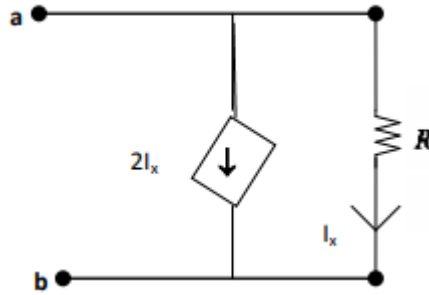
1. The defining equations of two circuits are as following:
 1. $y \frac{dy}{dt} + y = 1$ where $y = y(t)$ 2. $\frac{dx}{dt} + \cos x = 0$ where $x = x(t)$
 Select the correct statement from the following
 (a) circuit 1 is linear and circuit 2 is nonlinear (b) circuit 2 is linear and circuit 1 is nonlinear (c) both circuits are linear (d) both circuits are nonlinear
2. V-I characteristics of an electric circuit element is described in the following table. The circuit element is
- | | | | | | | | | |
|-----------------|---|----|----|----|---|---|---|---|
| Voltage (volts) | V | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
| Current (amps) | I | -1 | -1 | -1 | 0 | 1 | 4 | 9 |
- (a) Unilateral and passive (b) nonlinear and bilateral (c) linear and bilateral (d) nonlinear and active
3. The power consumed/generated in the voltage source (P_V) and current source (P_I) are (Negative sign for power generated)
- (a) $P_V = -30W$ $P_I = -12W$ (b) $P_V = -12W$ $P_I = -12W$
 (c) $P_V = -20W$ $P_I = +12W$ (d) $P_V = -30W$ $P_I = +12W$
- 
4. A box contains a circuit of resistances, dependent sources and independent sources. Only two terminals are available for external measurements. Following methods are suggested for measuring Thevenin's equivalent resistance of the circuit viewed from external terminals.
1. Open circuit voltage divided by short circuit current
 2. Connect a 1V voltage source at the external terminals. Find the input current using an ammeter. Thevenin's resistance is 1V divided by the input current
- (a) method 1 is correct (b) method 2 is correct (c) both the methods are correct (d) both the methods are wrong.
5. When a resistance of 0.5Ω is connected across the load terminals a maximum power of 50W is transferred. If a resistance of 1.5Ω is connected across the load terminals the power transferred to the load is
- (a) 16.667W (b) 18.75 W (c) 33.33 W (d) 37.5 W [5x2=10]

Name :

Roll No :

	a	b	c	d	
1	☐	☐	☐	☐	6
2	☐	☐	☐	☐	7
3	☐	☐	☐	☐	
4	☐	☐	☐	☐	
5	☐	☐	☐	☐	8

6. What is the equivalent resistance at terminals ab for the circuit given? $R = 3\ \Omega$ [3]



7. A graph of the circuit consists of 21 branches and 15 nodes. What is the order of the fundamental circuit matrix of the graph. [1]
8. When the nodal equations are formulated using fundamental cut set matrix, we can obtain the equations as $YV_t = J_n$ where Y is the product of 3 matrices and J_n is the product of two matrices. Draw the appropriate graph of the circuit and find these 5 matrices. R_2 branch should be a link. The voltage source is of 20 V and the current source of 2 A. $R_1 = 5\ \Omega$, $R_2 = 20\ \Omega$ and $R = 2\ \Omega$ [6]

